Brass harness: what connects to what

The whole harness as one list. 11 pins leave A1; 2 rails carry power and ground to everything; 2 resistors and 2 capacitors sit between. Rod names, lengths and heights match the step-by-step page, which is where the drawings and the build order are.

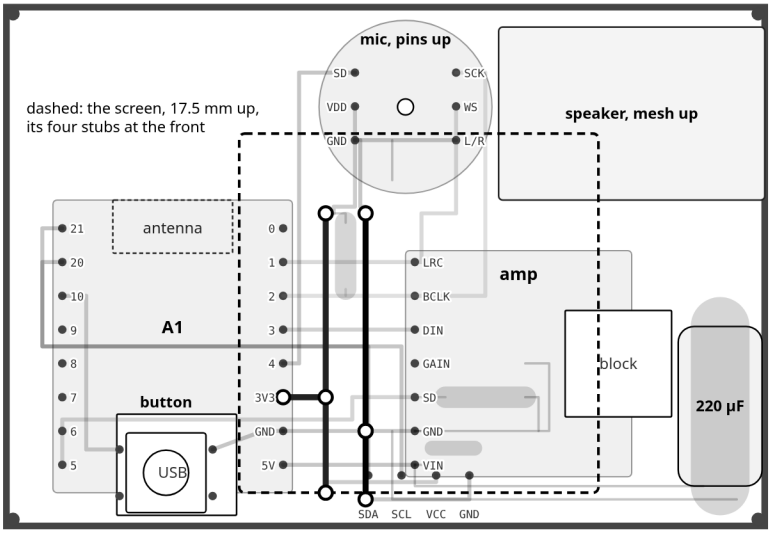
The two rails

A rail is a bare, straight 1 mm rod lying on the jig card during the build; in the finished device it hangs on its taps, lowest thing in the cage. It is the one wire that many parts share: every part's ground hooks onto the GND rail, every 3.3 V part hooks onto the 3V3 rail. A1 feeds each rail once; after that nobody runs a rod back to A1 for power.

RAIL	WHERE	FED BY	WHO HOOKS ON
3V3 rail	`x = 24, y = 2.5 → 23.5, on the card (21 mm)`	A1 3V3 stub, by the 3V3 tap	screen VCC, mic VDD, R3 (10 kΩ pull-up)
GND rail	`x = 27, y = 2.0 → 23.5, on the card (21.5 mm)`	A1 GND stub, by the GND rod and a 3 mm drop pin	amp GND (through the GND rod), screen GND, mic GND, mic L/R, button, R4, the 220 µF −

They lie in the channel, the 8.5 mm gap between A1 and the amp, 3 mm apart, 2 mm from the front edge, ending 1 mm short of the mic. The straight rods cross 1.5 mm above them. x is millimetres from the left wall, y from the front wall.

The two rails and the two pieces that feed them



front of the device (USB) · 10 px = 1 mm

Bold: the rails, the 3V3 tap and the GND drop pin. Everything else faded for position.

Every A1 pin that is used

A1 PIN	GOES TO	BY	LENGTH	HEIGHT	NOTE
5V	amp VIN	5 V rod	10 mm	rises to 5 mm	USB power for the amp only; the 220 µF hangs on this net
GND	amp GND, and the GND rail	GND rod + GND drop pin	14 mm	rises to 5 mm	one straight rod to the amp; the pin under it makes the rail ground
3V3	3V3 rail	3V3 tap	7 mm	rises to 4.7 mm	feeds the screen and the mic
GPIO4	mic SD	mic SD rod	32 mm	rises to 8 mm	sound in; pulled down inside A1 by the firmware (no resistor since 2026-10-14)
GPIO3	amp DIN	DIN rod	10 mm	rises to 5 mm	sound out
GPIO2	amp BCLK, then mic SCK	BCLK rod	38 mm	rises to 8 mm	one rod, two taps; R3 (10 kΩ) to 3V3 on this net
GPIO1	amp LRC, then mic WS	WS rod	33 mm	rises to 10 mm	one rod, two taps
GPIO5	amp SD	amp SD rod	43 mm	rises to 10.5 mm	amplifier on/off; R1 (4.7 kΩ) to GND on this net
GPIO10	button	button rod	19 mm	rises to 8.1 mm	the button's other side goes to A1 GND
GPIO20	screen SCL	SCL rod	53 mm	rises to 13.7 mm	I^2C clock
GPIO21	screen SDA	SDA rod	53 mm	rises to 13.7 mm	I^2C data

Unused and left as stubs: GPIO0, 6, 7, 8, 9. GPIO9 is the BOOT button, never an input.

Power and ground taps

PART PIN	TO	BY	LENGTH
screen GND	GND rail, front end	screen GND tap	23 mm
screen VCC	3V3 rail, front end	screen VCC tap	22 mm
mic GND	GND rail, back end	mic GND tap	11 mm
mic VDD	3V3 rail, back end	mic VDD tap	20 mm
mic L/R	mic GND stub (selects the left slot)	L/R link	8 mm
button, right legs	A1 GND stub	button GND rod	8 mm
amp GND	A1 GND stub	GND rod	10 mm

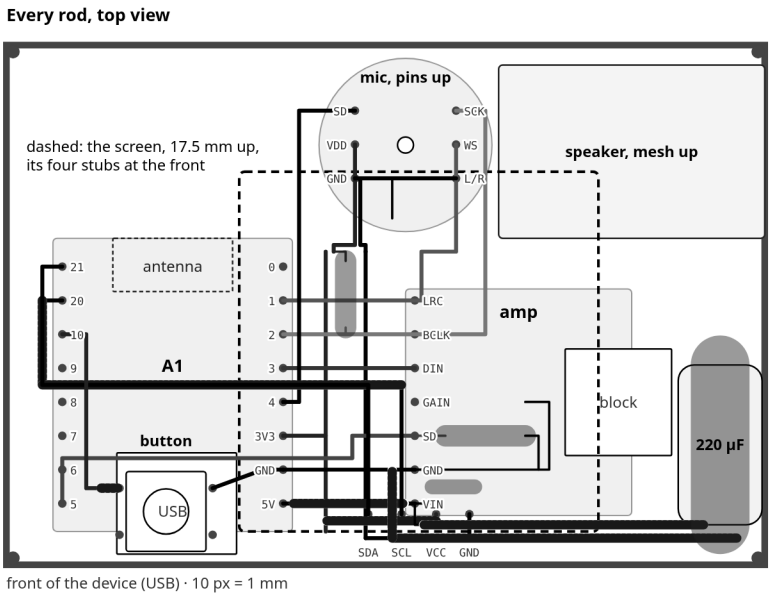
The six small parts

PART	BETWEEN	WHY	WHERE IT SITS
R1 4.7 kΩ (or the 1 kΩ you chose in Step 11)	amp SD stub ↔ amp GND stub	holds the amp off while A1 boots	a U lying over the amp board
~~R2~~ 100 kΩ	—	dropped 2026-10-14: GAIN open is 9 dB; fit 100 kΩ GAIN↔GND later for 15 dB	—
R3 10 kΩ	BCLK rod ↔ 3V3 rail	GPIO2 reads high at boot	in the channel, 8.5 mm up
~~R4~~ 100 kΩ	—	dropped 2026-10-14: the firmware enables GPIO4's internal pull-down	—
C1 220 µF, stripe to GND	5 V net ↔ GND rod	bass and Wi-Fi bursts don't sag the 5 V	lying in the right strip, leads sleeved along the floor
C2 100 nF	amp VIN stub ↔ amp GND stub	the fast part of the same job, right at the pins	flat on the amp board

Never connected

- **OUT+ and OUT−** go only to the speaker's own two leads. Never to GND, a rail, or the frame.
- **The frame** carries nothing. Every rail and rod stays 2 mm from it, with Kapton on the tubes.
- **5 V** reaches the amp only. The screen and mic live on 3V3.

All of it on one map



Front at the bottom. Dots are stubs, dotted stretches are heat-shrink, the dashed square is the screen floating above.

5 V GND 3V3

- WS (GPIO1)
- BCLK (GPIO2)
- DIN (GPIO3)
- mic SD (GPIO4)
- amp SD (GPIO5)
- button (GPIO10)
- SCL (GPIO20)
- SDA (GPIO21)

Sources. The net list is fast-track Steps 6–11 and the `CLAUDE.md` hardware contract; rod names, lengths and heights are generated from `edu/pages/gen/harness_model.py`, the same model as the step-by-step page. The circuit passed on the breadboard; the harness has not been built yet.